

ECON 311

Lecture 8: Solow Growth Model, part III

Solving for Equilibrium

- For any t , the aggregate economy in the Solow Model is characterized by five equations and five unknowns (K_t, L_t, I_t, C_t, Y_t):

$$\Delta K_{t+1} = \bar{s}F(K_t, L_t) - \bar{d}K_t \quad (\text{law of motion})$$

$$Y_t = \bar{A}K_t^{\frac{1}{3}}L_t^{\frac{2}{3}} \quad (\text{production function})$$

$$C_t + I_t = Y_t \quad (\text{resource constraint})$$

$$L_t = \bar{L} \quad (\text{labour force})$$

$$I_t = \bar{s}Y_t \quad (\text{allocation of resources})$$

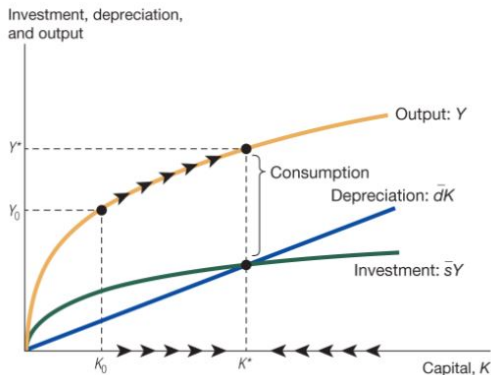
- For an initial level of capital, K_0 , we can solve for the endogenous variables in any time period.
 - Instead of solving the model for any time period, which requires a lot of algebra, we will only solve analytically for the steady state (ss) and provide graphical interpretation for other time periods.

Output and Consumption per capita

- An important variable in the Solow Model is the level of consumption at time t , which can be written as:

$$C_t = (1 - \bar{s})Y_t$$

The Solow Diagram with Output

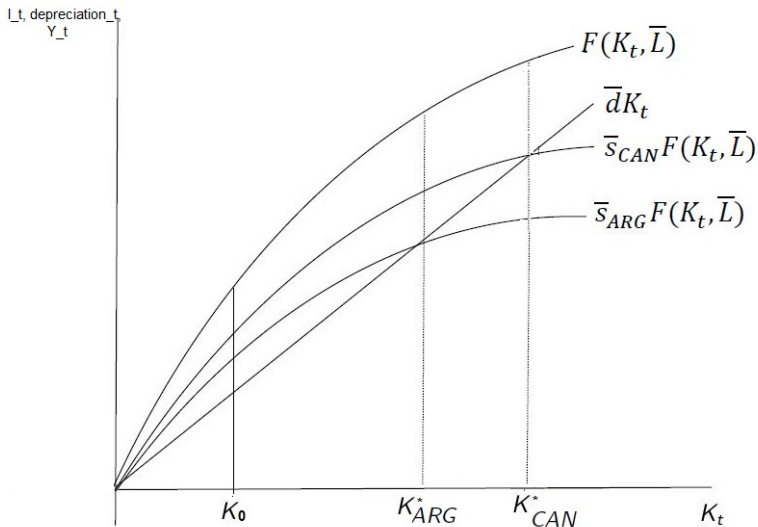


Output and Consumption per capita

- The resource constraint formalizes the trade-off an economy faces:
 - ▶ In order to grow faster and achieve higher production and higher consumption in the long run, initial consumption must be lower.
- We can see this point in our example of the Canadian and Argentine economies:

$$t = 0 \quad c_0^{ARG} = (1 - \bar{s}_{ARG})\bar{A} \left(\frac{K_0}{L} \right)^{\frac{1}{3}} > (1 - \bar{s}_{CAN})\bar{A} \left(\frac{K_0}{L} \right)^{\frac{1}{3}} = c_0^{CAN}$$

Output and Consumption per capita



Solving for Steady State

- To solve for the steady state equilibrium, we can reduce the equilibrium conditions to three equations and three unknowns:

$$\Delta K_{t+1} = 0 \quad \Rightarrow \quad K^* = \bar{L} \left(\frac{\bar{s}\bar{A}}{d} \right)^{\frac{3}{2}}, \quad k^* = \left(\frac{\bar{s}\bar{A}}{d} \right)^{\frac{3}{2}}$$

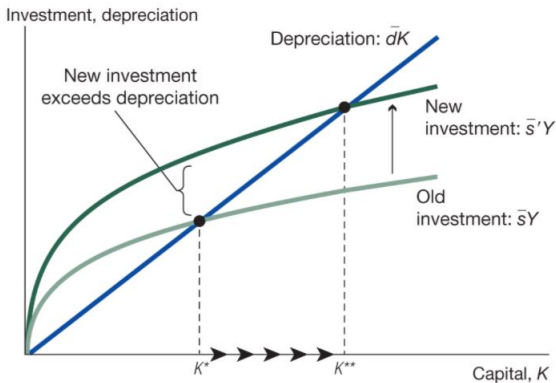
$$Y^* = \bar{A}(K^*)^{\frac{1}{3}} (\bar{L})^{\frac{2}{3}} \quad \Rightarrow \quad Y^* = \bar{L} \bar{A}^{\frac{3}{2}} \left(\frac{\bar{s}}{d} \right)^{\frac{1}{2}}, \quad y^* = \bar{A}^{\frac{3}{2}} \left(\frac{\bar{s}}{d} \right)^{\frac{1}{2}}$$

$$C^* = (1 - \bar{s})Y^* \quad \Rightarrow \quad C^* = (1 - \bar{s})\bar{L} \bar{A}^{\frac{3}{2}} \left(\frac{\bar{s}}{d} \right)^{\frac{1}{2}}, \quad c^* = (1 - \bar{s})\bar{A}^{\frac{3}{2}} \left(\frac{\bar{s}}{d} \right)^{\frac{1}{2}}$$

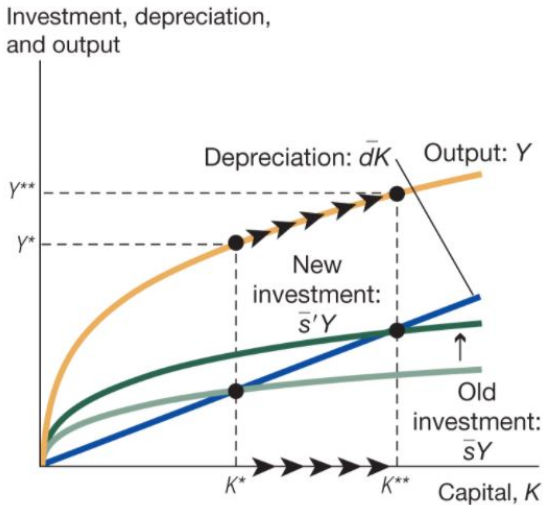
- We will also study how the system behaves out of the steady state without solving for endogenous variables.
 - ▶ This is important to understand differences in growth rates.

An increase in savings rate $\bar{s}$

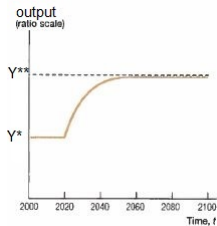
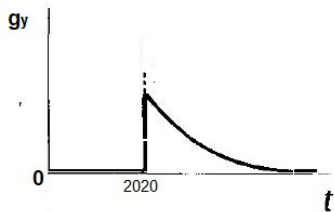
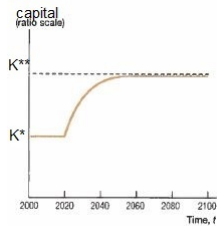
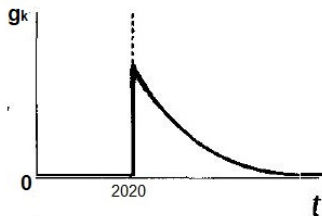
- Assume that the economy is in its steady-state equilibrium and in 2020, there is an increase in savings:



An increase in savings rate $\bar{s}$

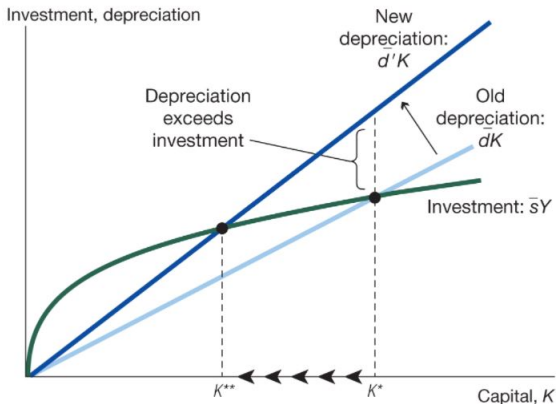


Transition Dynamics

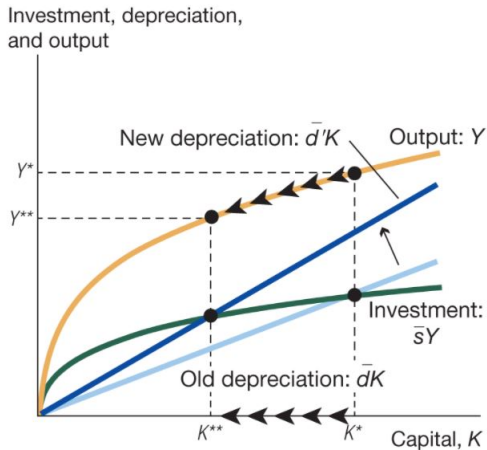


An increase in depreciation rate $\bar{d}$

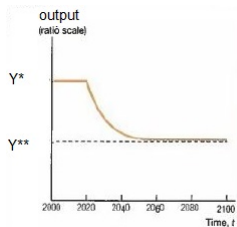
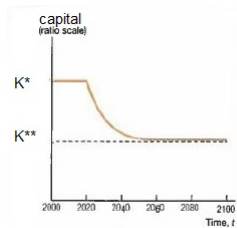
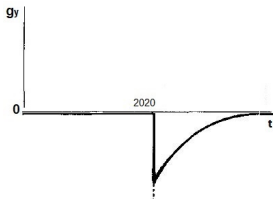
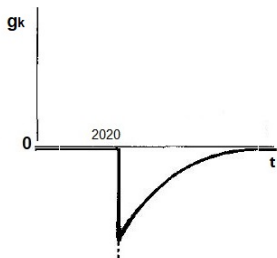
- Assume that the economy is in its steady-state equilibrium and in 2020, there is an increase in depreciation rate:



An increase in depreciation rate $\bar{d}$



Transition Dynamics

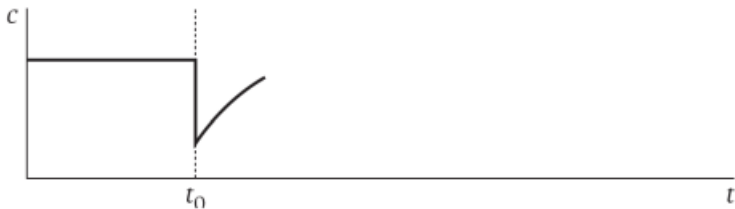


Consumption

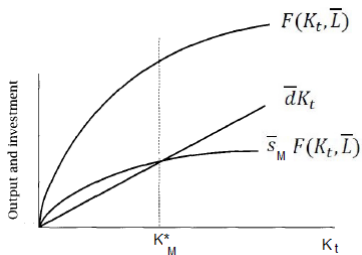
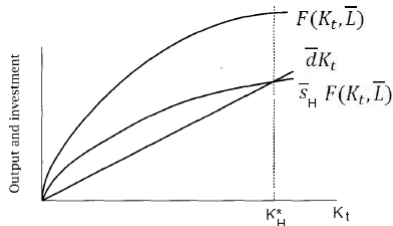
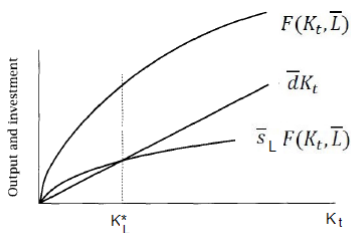
- As we showed on the ARG vs. CAN example, higher savings lead to higher GDP (per capita).
 - ▶ Households' welfare does not depend on output per capita but on consumption per capita.
- A higher level of savings, *ceteris paribus*, results in ...
 1. an increase in GDP (per capita) increases consumption (per capita)
 2. an increase in the share of output invested decreases consumption (per capita)
- It turns out that in the Solow Model there is a saving rate that maximizes consumption (per capita) in the steady-state
 - ▶ The amount of capital per capita maintained by this saving rate in equilibrium is called the *golden-rule* level of capital.

An increase in savings rate $\bar{s}$

- Assume that the economy is in its steady-state equilibrium and there is an increase in savings:



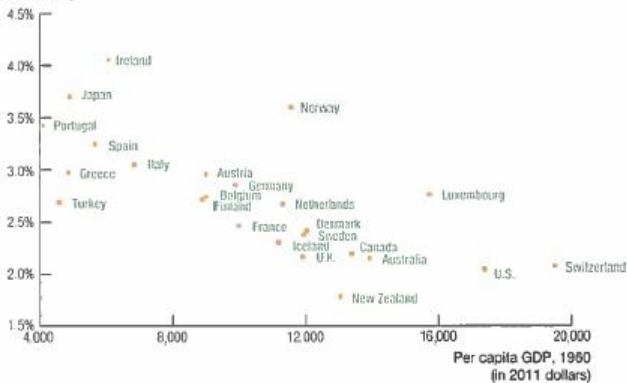
Consumption and Savings in the steady state



Convergence with similar y^{ss}

Growth Rates in the OECD, 1960–2014

Per capita GDP growth
(1960–2014)



Conclusion– Simple Solow Model

- Provides an explanation as to why growth rates are different across countries.
 - ▶ Countries have different rates of capital accumulation
 - ▶ It is only useful in explaining differences among a homogeneous set of countries (with similar steady state)
- Explicitly model the trade-off between current consumption and future growth
 - ▶ Reducing current consumption increases savings and future output
 - ▶ Too much savings can lead to a decrease in the steady state level of consumption
- A big failure of the simple Solow Model is that ...
 - ▶ there is no economic growth in the long run.
 - ▶ it only provides an explanation for different y^{ss} through different k^{ss}